From variational Monte Carlo to superfluid hydrodynamics in microwave-shielded Bose Einstein condensates

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We show that, within the manifold where the Jastrow pair factor is frozen to its ground-state form, time-dependent variational Monte Carlo (tVMC) based on a Jastrow–Feenberg wavefunction maps exactly onto irrotational superfluid hydrodynamics. The only microscopic input is an equation of state (EOS) extracted from a single variational Monte Carlo calculation of the uniform system via the local-density approximation, which yields a density-dependent chemical potential $\mu(\rho)$ for a generalized Gross–Pitaevskii equation (GPE). For a trapped microwave-shielded dipolar Bose gas, this reduced description reproduces the full tVMC breathing dynamics for $N = 8$–128 with below-1% frequency error and $\sim 2\%$ amplitude error, while lowering the computational cost by more than an order of magnitude. This framework provides a rigorous, first-principles route to coarse-grain highly correlated wavefunctions into predictive macroscopic hydrodynamics, applicable broadly to molecular gases, quantum droplets, and supersolids.

Introduction.—Microwave-shielded polar molecules have emerged as a promising platform for quantum many-body physics, with recent experiments achieving Bose–Einstein condensation [1], observing shielded collisions [2, 3], and probing dipolar many-body effects [4–6]. A central theoretical challenge is to predict nonequilibrium collective dynamics—breathing oscillations, quenches, and expansion—from the microscopic interaction, which combines long-range dipolar attraction ($\sim C_3/r^3$) with short-range repulsive shielding ($\sim C_6/r^6$).

Mean-field Gross–Pitaevskii (GP) theory is computationally efficient but reduces this interaction to a contact nonlinearity, thereby missing the anisotropic and density-dependent correlations generated by the shielding mechanism. At the opposite end, time-dependent variational Monte Carlo (tVMC) [7, 8] can treat realistic interactions within a correlated many-body wavefunction, but its stochastic sampling at every time step severely limits accessible system sizes and evolution times.

This Letter introduces a general theoretical framework that coarse-grains a highly correlated microscopic wavefunction into a macroscopic hydrodynamic theory, retaining the exactness of tVMC while recovering the efficiency of the GPE. While demonstrated here for shielded molecular gases, this approach provides a rigorous, first-principles alternative to purely phenomenological extended GPEs for describing collective dynamics in strongly correlated fluids like quantum droplets, supersolids, and interacting Rydberg gases. Our central premise relies on a fundamental separation of scales: short-range pair correlations relax much faster than the global density redistribution. Under this physical condition, once the two-body Jastrow factor u_2 is frozen, the time-dependent variational principle (TDVP) maps exactly onto irrotational superfluid hydrodynamics. We first verify from full tVMC that u_2 remains nearly stationary during the quench, then derive a generalized GPE governed by a uniform-system EOS via the local-density approximation (LDA). The resulting dynamics reproduces the full tVMC benchmarks for $N = 8$ and $N = 128$, easily extending to experimentally relevant large-N regimes beyond direct tVMC.

Model and variational dynamics.—We consider N bosons in a cylindrically symmetric harmonic trap with Hamiltonian

$$\hat{H}(t) = \sum_{i=1}^{N}\left[-\tfrac{1}{2}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \qquad (1)$$

where $V_{\rm tr} = \frac{1}{2}[\omega_\rho(t)^2\rho^2 + \omega_z^2 z^2]$ (natural units $\hbar = m = 1$) and the two-body interaction models microwave-shielded dipolar molecules:

$$v(\mathbf{r}) = \frac{C_3}{r^3}(3\cos^2\theta - 1) + \frac{C_6}{r^6}(\sin^4\theta + w_{12}\sin^2\theta\cos^2\theta). \quad (2)$$

This effective potential, derived from Floquet scattering theory for dual-microwave-dressed molecules [9], is parametrized by coefficients C_3, C_6, and w_{12} that are determined by the microwave field configuration. In a realistic experiment these are fixed by the dressing scheme; here, for demonstration, we choose $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units. We study a quench $\omega_\rho : 1 \to 2$ at $t = 0$ that excites radial breathing modes.

The many-body wavefunction is a truncated Jastrow–Feenberg [10, 11] ansatz:

$$\Psi(\mathbf{X}, t) = \exp\Big[\sum_i u_1(\mathbf{r}_i, t) + \sum_{i<j} u_2(r_{ij}, \hat{\mathbf{r}}_{ij})\Big], \qquad (3)$$

where $u_1(\mathbf{r}, t)$ is a complex, time-dependent one-body field and u_2 encodes two-body correlations. In cylindrical geometry we use a separable parametrization for u_1, while u_2 is expanded in partial waves as

$$u_2(r, \cos\theta) = u_{\rm cusp}(r) + \sum_{\ell=0,2,4} f_\ell(r)\, P_\ell(\cos\theta), \qquad (4)$$

where $u_{\rm cusp}(r)$ captures the short-range scattering boundary condition and each f_ℓ is represented by Gaussian basis functions [12]. For the parameters studied here, the $\ell = 4$ channel is negligible and the $\ell = 2$ anisotropy is weak, so the pair correlation is close to isotropic.

Figure 1 shows the quality of the optimized pair correlations. The leading-order hypernetted-chain (HNC) approximation to the pair distribution function, $g_2^{\rm HNC}(r, \theta) = e^{2u_2(r,\cos\theta)}$ [13], reveals a correlation hole at short range [Fig. 1(a)], reflecting the interplay between dipolar attraction along $\theta = 0$ and the shielding repulsion. The pair correlation is nearly isotropic, consistent with the subdominant role of f_2.

The TDVP [14, 15] projects the Schrödinger equation onto the variational tangent space. Defining logarithmic derivatives $O_\alpha = \partial_{\theta_\alpha} \ln \Psi$ and the local energy $E_L = (\hat{H}\Psi)/\Psi$, the equations of motion read [7]

$$\sum_\beta S_{\alpha\beta}\, \dot{\theta}_\beta = -i\, F_\alpha, \qquad (5)$$

where the overlap matrix and force vector are

$$S_{\alpha\beta} = \langle \Delta O_\alpha^* \, \Delta O_\beta \rangle_t, \qquad (6)$$
$$F_\alpha = \langle \Delta O_\alpha^* \, \Delta E_L \rangle_t, \qquad (7)$$

with $\Delta A = A - \langle A \rangle_t$ denoting connected fluctuations. Full Monte Carlo and convergence details are given in Ref. [12].

Frozen-u_2 reduction.—The physical justification for freezing the Jastrow pair factor originates from the separation of time and length scales. Short-range pair correlations governed by $v(r)$ quickly equilibrate, whereas the large-scale density profile evolves smoothly, driven by the trap quench. Indeed, our full tVMC confirms this physical picture: during the quench, the optimized two-body parameters change by less than 1% of their ground-state values, while the one-body field changes by order unity. Figure 1(b) already shows that a frozen-u_2 tVMC evolution accurately tracks the full tVMC dynamics, and Fig. 2 visualizes the near-stationarity of the pair correlation itself. This motivates analytically restricting the variational manifold to a fixed u_2.

When the variational degree of freedom is the continuous field $u_1(\mathbf{r}, t)$, the TDVP logarithmic derivative becomes the microscopic density operator:

$$O(\mathbf{r}; \mathbf{X}) = \frac{\delta \ln \Psi}{\delta u_1(\mathbf{r})} = \sum_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i) = \hat{\rho}(\mathbf{r}). \qquad (8)$$

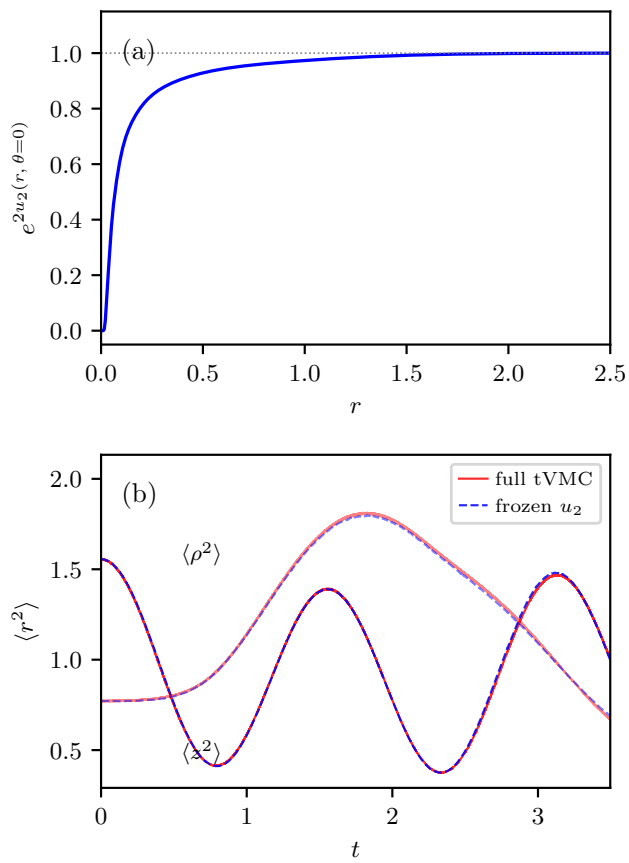

FIG. 1. (a) Leading-order HNC pair correlation $e^{2u_2(r,\theta=0)}$ for the N=8 ground state, showing a correlation hole at short range due to the shielding repulsion. (b) Breathing dynamics after the quench $\omega_\rho : 1 \to 2$ for N=128: $\langle \rho^2 \rangle(t)$ and $\langle z^2 \rangle(t)$ from full tVMC (red) and frozen-u_2 tVMC (blue dashed), validating the frozen-u_2 approximation.

Because u_2 is real and frozen, the many-body phase is purely one-body and irrotational: $\mathbf{v} = \nabla(\mathrm{Im}\, u_1)$. No backflow terms arise.

Decomposing $u_1 = a + i\theta$ (real amplitude and phase), the TDVP action reduces to the standard hydrodynamic form [12]. Within this frozen-u_2 manifold the reduction is exact and yields the continuity and Euler equations:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \qquad (9)$$
$$\partial_t \mathbf{v} + \nabla\Big(\tfrac{1}{2} v^2 + V_{\rm tr} + \mu_{u_2}[\rho]\Big) = 0, \qquad (10)$$

where $\mu_{u_2}[\rho] = \delta E_{\rm int}[\rho; u_2]/\delta\rho$ is the effective chemical potential determined by the frozen correlations.

To obtain a closed local theory, we further invoke the local-density approximation (LDA). The internal energy then takes the form $E_{\rm int} = \int \rho\, \varepsilon(\rho)\, d^3 r$, where $\varepsilon(\rho)$ is the energy per particle of the *uniform* system at density ρ with the same frozen u_2. The chemical potential becomes

a local function:

$$\mu(\rho) = \frac{d[\rho\varepsilon(\rho)]}{d\rho} = \varepsilon + \rho\varepsilon'. \tag{11}$$

This is structurally identical to GP theory, except that the usual contact coupling gn is replaced by $\mu(\rho)$, which encodes the full effect of the realistic, anisotropic interaction through the frozen correlations.

The near-stationarity of u_2 during the quench is visualized directly in Fig. 2, which shows the change in the leading-order HNC pair correlation Δe^{2u_2} at four time snapshots. The variation remains below ~ 0.03 throughout (compared to peak values ~ 1), confirming that the pair correlations are effectively frozen.

Equation of state and generalized GPE.—We extract $\varepsilon(\rho)$ from a one-time VMC calculation on the uniform system (periodic boundary conditions) with u_2 frozen at its trapped ground-state values (computational details are given in [12]). Over the relevant density ranges, we fit the results to a polynomial

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + \cdots, \tag{12}$$

yielding the chemical potential $\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + \cdots$.

This robust framework offers a striking contrast to traditional standard extended GP theories, where beyond-mean-field physics is typically incorporated by adding a phenomenological Lee–Huang–Yang (LHY) term. Since the original LHY correction is strictly derived from the quantum fluctuations of a uniform ground-state energy, directly asserting it as a local dynamical potential in the GPE lacks a rigorous first-principles footing. Our approach, in contrast, structurally derives the extended equation natively from the exact TDVP mapping combined with the corresponding EOS. Furthermore, while the classic LHY correction scales as $\rho^{3/2}$ for contact interactions (and introduces analogous generic non-analytic terms for bare dipoles), the precise density dependence of beyond-mean-field effects is intimately governed by the specific system EOS. For our microwave-shielded dipolar configuration, the uniform-system VMC yields a well-behaved EOS matching Eq. (12).

In this expansion, the leading term $2a_1\rho$ corresponds to a *renormalized two-body coupling* $g_{\rm eff} = 2a_1$, which absorbs the effect of the frozen pair correlations into an effective contact interaction. The next-order term $3a_2\rho^2$, however, has the structure of an *effective three-body interaction* with coupling strength $g_3 = 3a_2$. This is not a genuine microscopic three-body force; rather, it emerges from integrating out the density dependence of the two-body correlations rigorously encoded in the frozen u_2.

For the shielded dipolar interaction studied here, we find $g_3/g_{\rm eff}^2 \sim 0.1$–$0.2$, indicating that the three-body term provides a 10–20% correction at the equilibrium density. For $N = 128$, a quadratic fit with $a_1 \approx 0.111$ and $a_2 \approx 0.001$ describes the relevant density range safely and remains stable ($d\mu/d\rho > 0$). The EOS is therefore close to linear at high density, reflecting the long-range character of the shielded dipolar interaction.

The VMC-derived chemical potential is shown in Fig. 3, confirming the nearly linear dependence $\mu \approx 2a_1\rho$ with a small quadratic correction.

The resulting generalized GPE,

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + g_{\rm eff}|\psi|^2 + g_3|\psi|^4\Big]\psi, \tag{13}$$

is therefore a *two-plus-three-body* GPE whose couplings are fixed microscopically by VMC rather than by phenomenology. This provides a first-principles realization of the extended GPE forms discussed in the contexts of quantum droplets [16] and dipolar condensates [17, 18].

GPE solver.—We solve the generalized GPE

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + \mu(|\psi|^2)\Big]\psi, \tag{14}$$

where $\psi = \sqrt{\rho}\, e^{i\theta}$ and the Laplacian provides both quantum pressure and flow kinetic energy. The equation is discretized on a cylindrical (r, z) grid exploiting azimuthal symmetry, evaluated with a fourth-order finite-difference Laplacian and time-integrated using a Runge–Kutta scheme under a dispersive CFL condition [12].

The initial state is obtained by imaginary-time propagation to self-consistency: $\psi \to \psi - d\tau\, \hat{H}\psi$, normalized to N particles at each step, until the chemical potential converges to $\delta\mu < 10^{-8}$.

Validation against tVMC.—We benchmark the generalized GPE against frozen-u_2 tVMC at both weak and moderate interaction strength. For $N = 8$, the GPE ground state reproduces the tVMC moments to 0.7% and the energy to 0.15% (Table S1 [12]). The breathing frequency is $\omega_{\rm br} = 4.19 \approx 2\omega_\rho$ in both methods, and the oscillation amplitudes of $\langle\rho^2\rangle$ and $\langle z^2\rangle$ agree to within 1%. The quantum-pressure term is essential in this regime because the interaction energy is only $\sim 3.5\%$ of the total.

For $N = 128$, where the interaction energy reaches $\sim 21\%$ of the total, the same EOS-based GPE remains accurate. The ground-state moments agree with tVMC to 0.35% for $\langle\rho^2\rangle$ and 1.05% for $\langle z^2\rangle$, while the total energy differs by 2.8%, consistent with a residual LDA error at higher density. More importantly, the dynamical benchmark in Fig. 4(a) is essentially unchanged: $\omega_{\rm br} = 4.19$ in both approaches and the $\langle\rho^2\rangle(t)$ amplitude agrees to $\sim 2\%$.

The computational gain is substantial. For $N = 128$, the GPE evolution including imaginary-time state preparation takes ~ 70 minutes on a single CPU core, compared with ~ 6 hours on a modern GPU for the full

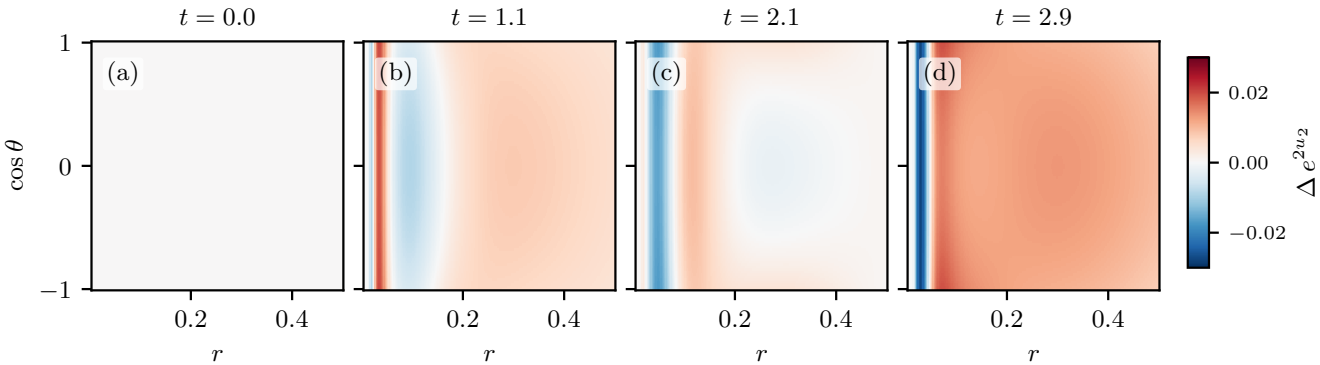

FIG. 2. Change in the leading-order HNC pair correlation, $\Delta e^{2u_2} = e^{2u_2(t)} - e^{2u_2(0)}$, during breathing dynamics for N=128. The pair correlations remain nearly frozen ($|\Delta e^{2u_2}| \lesssim 0.03$) while the one-body density undergoes order-unity changes.

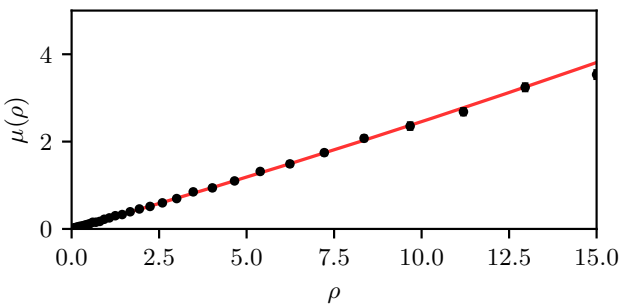

FIG. 3. Chemical potential $\mu(\rho) = d(\rho\varepsilon)/d\rho$ from uniform-system VMC (circles) for C_3=10^{-3}. The line shows the polynomial fit $\mu = 2a_1\rho + 3a_2\rho^2$ used in the GPE.

tVMC. The reduction is larger still in total floating-point cost, since the stochastic tVMC evolution requires fresh Monte Carlo sampling at every Runge–Kutta stage.

Large-N prediction.—Having benchmarked the reduction, we use it in a regime inaccessible to direct tVMC. With the same interaction ($C_3 = 10^{-3}$), the generalized GPE predicts sustained breathing oscillations for $N = 8192$ over six trap periods [Fig. 4(b)], with excellent energy conservation ($|\delta E/E| < 2 \times 10^{-4}$). This illustrates the immense practical value of the approach: it seamlessly scales up to $N \sim 10^4$, matching the experimentally relevant molecule numbers in recent milestone breakthroughs such as those by Bigagli et al. [1]. Once the EOS is determined from a single VMC calculation, the complex correlated many-body quench problem is completely reduced to a deterministic hydrodynamic evolution.

Discussion.—The success of the frozen-u_2 hydrodynamic reduction rests on a fundamental physical separation of scales: short-range pair correlations governed by interactions equilibrate much faster than the global macroscopic density redistribution, keeping u_2 nearly stationary.

Two features fundamentally distinguish the present framework from phenomenological extended GP theories. First, the density-dependent chemical potential is derived directly from the microscopic correlated many-body wavefunction, completely capturing the true geometric complexities of $v(r)$, rather than being heuristic. Second, the hydrodynamic equation itself is not postulated: it follows directly from the TDVP once the dynamics is *exactly* projected onto the frozen-u_2 manifold. In this sense, the approximation is structurally transparent and diagnosable: its validity can always be checked a posteriori by monitoring the time dependence of u_2.

This rigorous construction clarifies the true mathematical origin of higher-order nonlinearities. The effective three-body term $g_3|\psi|^4$ in Eq. (13) is not inserted ad hoc; it acts as a low-energy manifestation of the density dependence of correlated two-body interactions. This makes the method a powerful, first-principles correct route to extended GPEs for dipolar and molecular systems, bypassing the ambiguities around hard-patching the LHY quantum fluctuation energies into time-dependent local fields.

The main quantitative limitation visible here is the $\sim 3\%$ energy offset at $N = 128$, which points to gradient corrections beyond the LDA [19]. More generally, the present reduction should hold when pair correlations remain slaved to the evolving density and the flow stays irrotational. It can then be extended naturally to other collective modes, quench protocols, and trap geometries relevant to current molecular-gas experiments [1, 20].

Conclusion.—We have demonstrated that correlated many-body dynamics from tVMC maps exactly onto superfluid hydrodynamics within the frozen-u_2 manifold, with the required equation of state obtained from a single uniform-system VMC calculation. The resulting generalized GPE reproduces the breathing dynamics of microwave-shielded dipolar bosons from $N = 8$ to $N = 128$ with below-1% frequency error and $\sim$ 2% amplitude error, while providing access to large-

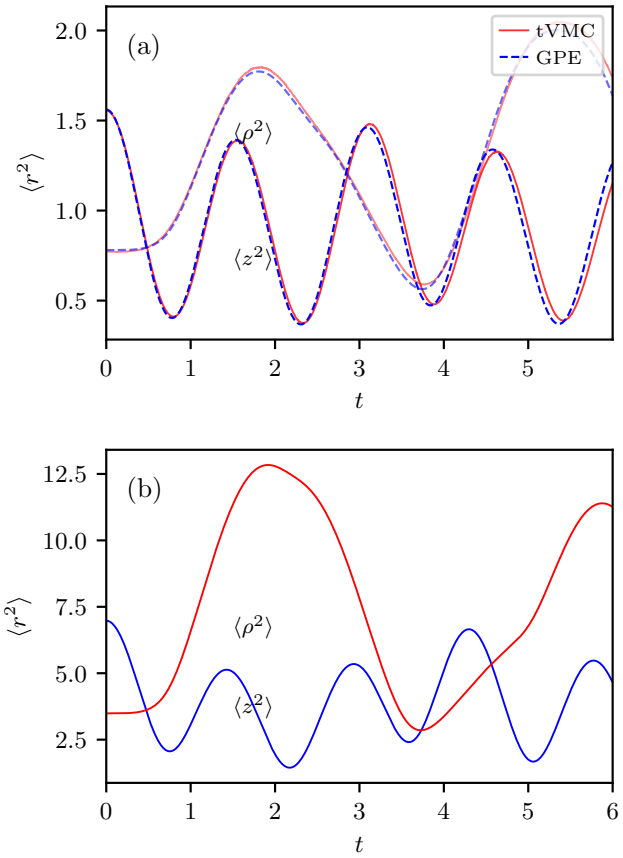

FIG. 4. GPE dynamics from the VMC-derived EOS. (a) N=128 validation: $\langle \rho^2 \rangle(t)$ and $\langle z^2 \rangle(t)$ from frozen-u_2 tVMC (red) and GPE (blue), showing identical breathing frequencies within numerical resolution. (b) N=8192 prediction with the same interaction (C_3=10^{-3}), extrapolating to experimentally relevant system sizes inaccessible to direct tVMC.

N regimes beyond direct tVMC. Beyond establishing a practical bridge between microscopic physics and predictive hydrodynamics for molecular gases, the "frozen-Jastrow" hydrodynamic concept introduced here establishes a powerful methodology. It has the potential to enable first-principles modeling of non-equilibrium dynamics in a wide array of quantum fluids governed by complex long- and short-range correlations, including quantum droplets, supersolids, and Rydberg atomic gases.

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